

Highlight the profile for which a name is to be entered or modified and press the Modify button. This will cause a window to appear allowing the required name to be entered using the button pad.

Once the name is displayed against the required profile the mode can be exited by pressing the 'X' in the top right corner of the Touchscreen Configuration window.

To exit this configuration window and return to the overview screen press the **Overview** button.

5.4.13 Cleaning the Touchscreen



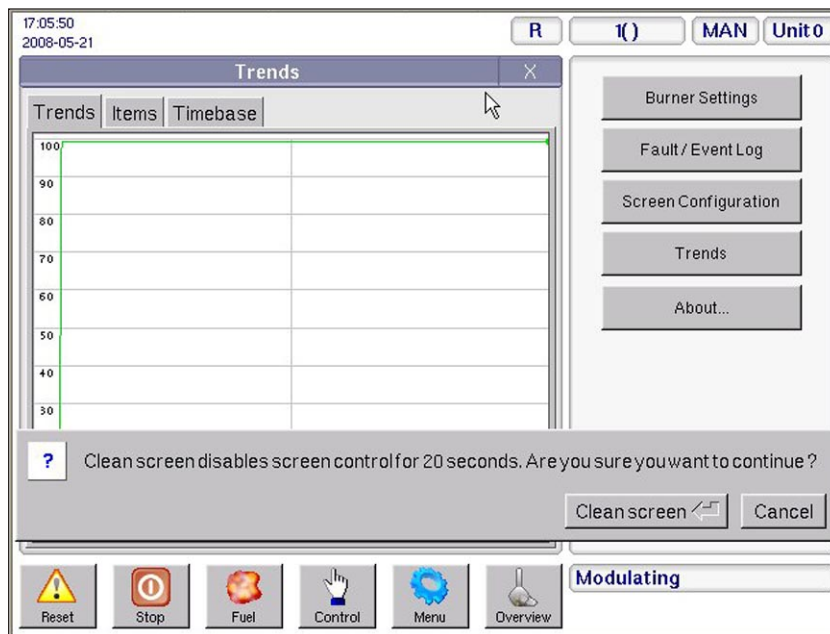
WARNING

To prevent possible damage to the touchscreen, ensure the correct 'fluid' is used when cleaning the screen.

Before attempting to clean the screen ensure the clean screen function is enabled to ensure operation of the burner is not affected

Once the 'Clean Screen' function is selected the display keys will be unavailable for 20 seconds

Before attempting to clean the screen it is essential to select the **Clean Screen** function to prevent any pressure applied to clean the touchscreen being 'seen' as a 'button press'. The function can be accessed by selecting the **Menu** button at the base of the touchscreen and then, selecting **Clean Screen** from the list of functions available.



When cleaning the touchscreen ensure the correct LCD screen cleaner is used. Use of any other cleaning fluid or water may damage the screen.

There are designated commercial cleaning solutions on the market for touchscreens only. Also, these products are recommended to be used only with a very soft cloth. Please note the directions and the warnings on the product.



CAUTION

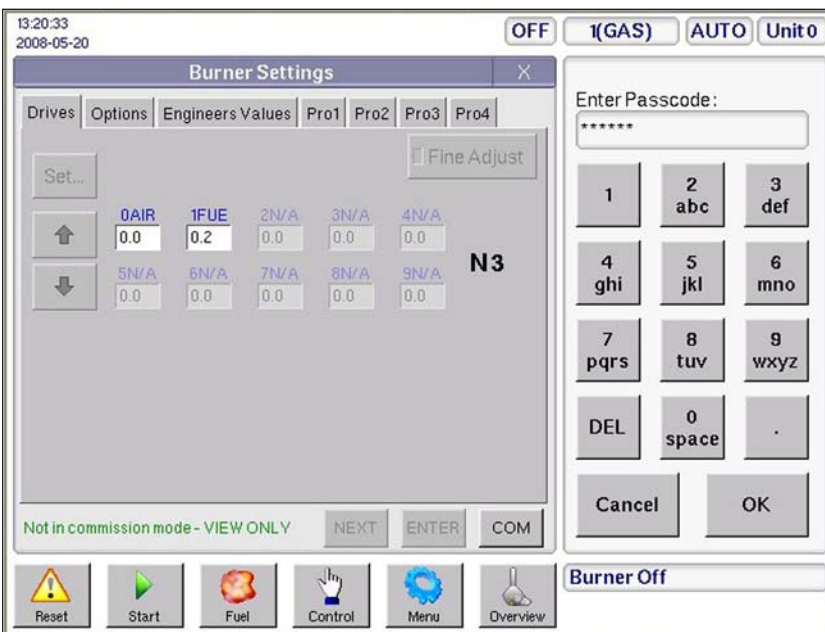
- Do NOT use any ammonia-based window cleaner. These chemical cleaners can ruin the touchscreen surface.
- Do NOT use any abrasive rags, towels or paper towels. An abrasive towel can scratch the touchscreen.

5.4.14 Touchscreen Commission mode.

To allow Option Parameters or Drives to be adjusted it is necessary to enter the relevant access passcode.

To enter the passcode press **MENU**, then **BURNER SETTING**, then the **COM** button. This will prompt the keypad pane to appear, allowing the relevant passcode to be entered using the soft keypad and confirmed by pressing the **OK** button.

NOTE: The Supplier's Passcode (LV3) is six digits. Enter ALL, including any zeros, followed by the **OK** button.

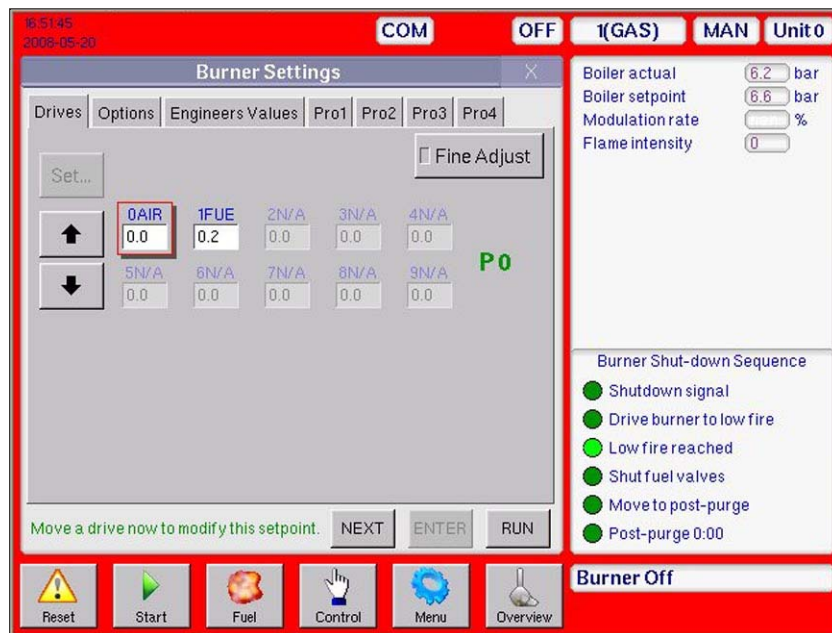




CAUTION

- Use extreme care while commissioning the system. While operating in commissioning mode the safety of the system is the sole responsibility of the commissioning engineer.
- Ensure a pre-purge position is entered for the relevant drives.
- Incorrect positioning of either fuel or air could cause a hazardous situation to occur.
- If the servo-motor positions have been up-loaded it is essential that the combustion is verified at each firing position to ensure a hazardous condition cannot occur, and for the data to be stored in the control.
- It is recommended that the close position for each servo motor is re-entered as part of the burner service regime, to compensate for wear in the servo-motor micro-switch during operation.
- Once all profile positions have been adjusted/entered it is essential that all profile points are checked to verify that the fuel air ration is acceptable for the appliance being controlled.

Once the correct passcode is entered the unit will enter commissioning mode and allow the option parameters and profile points to be adjusted, the text on the **COM** button will change to **RUN**. To warn the 'user' that the unit is in commissioning mode the overall display background changes to red and COM is displayed top-center on the screen



To enter or adjust Drive positions it is necessary to select the relevant Drive to be set, and then use the up/down arrows to move the drive to the required position. Set all the drive positions for each profile point before using the **ENTER** button to store the values and move to the next profile point (if already entered) then repeat the procedure for this point and any others required up to high fire.



NOTE: When adjusting servo motor positions, it may require the use of the "Fine Adjust" button to get the motor to the exact spot you desire.

Once the required values have been entered or changed the mode can be exited by pressing **RUN** followed by **ENTER**, the 'X' at the top right corner of the Burner Adjust window will be grayed out and inoperative.

5.5 Using option set mode

- Use the UP/DOWN keys to select the Option parameter to be adjusted.
- Use the Scroll (< >) keys to navigate between Option numbers and the Option values.
- Use the UP/DOWN keys to change the value of the option parameter. When the correct value is selected then press the ENTER key to store the new value.
- Set or adjust all of the parameters as required by moving between the parameters and adjusting as described above.

5.5.1 Option parameter list



CAUTION

- Use extreme care when entering option parameters. Incorrect data entry could cause a hazardous situation to occur.
- It is possible that additional parameters may be present in the unit depending on the application and/or supplier.

Note: Option parameters are marked with LV1, LV2, and LV3, which indicate what level passcode is required. If, when using the supplier passcode (LV3), it is not possible to adjust the value of an option parameter, then the burner must be turned off to make the adjustment.

Option 00.1 - Site passcode (0 - 999) DEFAULT VALUE is 154. LV3

This is a three-digit passcode that will allow the site engineer or end-user to enter option set mode and adjust a limited range of option parameters (those not marked with 'LV2' or 'LV3' in this section). This passcode can be zero, in which case the user only needs to press the COM / ENTER key twice to enter option set mode with limited access.

Option 00.2 - Serial communications control address (0 - 15) LV3

If the control is to be connected to other equipment via the serial communications interface, it must be given a unique address using this option parameter. Additionally if more than one control unit is connected on the CAN bus (for example to share a display), the units must all have unique addresses BEFORE THEY ARE CONNECTED TO THE SAME CANbus.



Option 00.3 - Reset hours run (0 - 1) LV3

This option parameter allows the HOURS RUN display for all fuels to be reset to zero. To perform a reset, set this option parameter to **1** and leave option set mode. When the burner begins to modulate, the HOURS RUN display will be reset and the option parameter automatically returns back to **0**.

Option 00.4 – User program select (0 / 1) LV1

This parameter may not be available. Where it is available, it allows the user to de-select the Fireeye's standard modulation program, and select a user program. The user program may be modified using PC software to change the controls behavior for non-standard applications.

The programmable system only allows access to non-safety related functions such as modulation, enabling / disabling burner start-up and other similar functions. Only the user program can be modified, and Fireeye's program can be re-selected at any time by changing this parameter back to zero.

Option 00.4 value	Meaning
0	Run standard manufacturer modulation program. Controls PID, control limits, and some Input / Output settings (see option parameters 20.0 to 29.9)
1	Run user program, if available.

Option 00.5 – Alternate Language select (0 / 1) LV1

DEFAULT: 0

This parameter may not be available. Where it is available, it allows the user to change the language used for the display. Depending on spare memory and customer requirements, this option may only change the text displayed to users in RUN mode, and not in COMMISSION mode – or all text can be changed to a second language.

Table below is for basic display. Language selection on the touchscreen is handled on the touchscreen directly.

Option 00.5 value	Meaning
0	Operate with standard language (usually ENGLISH).
1	Operate with alternate language, at least in RUN mode – but possibly in other modes depending on specification of the control.

Option 00.6 – Reserved

Option 00.7 – Modbus device address (1 – 99) LV3

Default value is 0= OFF

This option sets the Modbus device address for the controller and must be unique for each device on the bus.

NOTE: Only available with NXDBMB (Modbus) or NXDBVSD (variable speed drive) daughterboard installed.



Option 00.8 – Modbus communications speed (0 – 3) LV3

Default value 0

NOTE: Only available with NXDBMB (Modbus) or NXDBVSD (variable speed drive) daughterboard installed.

This option sets the Modbus communications speed in each controller as follows:

Option 00.8 value	Speed, bits/s
0	9600
1	4800
2	9600
3	19200

Option 01.0 – Power Up Option (0 - 30) LV3

DEFAULT 15 (after Sept 2011)

This option can be used to modify the behavior of the control at power-up.

Option 01.0 value	Meaning
0	Normal power-up. If no faults and fuel / burner selected, the burner will start.
1	Lockout. The burner will always lockout after a power-up. An F75 will be generated which will require manual intervention to clear.
2 – 30	The control will power-up as normal, but wait for 2 to 30 seconds before checking the digital inputs for fault conditions. This may be useful to prevent nuisance lockouts at power-up due to water levels and plant interlocks. The control will remain in safety shutdown (but not locked out) for this time.

Option 01.1 – Keyboard Auto/Manual enable (0 - 1) LV3 [Engineer's Key 33]

DEFAULT: 1

It is possible to enable/disable the auto/man facility from the keyboard. Selecting a 0 will disable the operation of the 'auto/man' function from the keyboard. Selecting a 1 will enable the operation of the 'auto/man' function from the keyboard.

Option 01.2 – Mute/Reset Input enable (0 - 2) LV3

The Mute/Reset function is available via Comfire 2, on the keyboard and can also be selected to be from a digital input. Selecting a non-zero value will enable the operation of the 'Mute/Reset' function from the corresponding digital input. It should be ensured that the digital input selected is not used for any other function (option parameters 1.x, 16.x and 18.x, 20.x). MUST be connected as shown in section 3.6.11.

Please note any key/button providing a fault mute function must be mounted local to the burner.

Number entered in parameter 01.2	Mute/Reset Function	Digital inputs used
0	Comview & Keyboard only	NONE
1	Input 1	Input 1, PA5 to PA11 Low Voltage
2	Input 2	Input 2, PA6 to PA11 Low Voltage
3	Input 3	Input 3, PA7 to PA11 Low Voltage
4	Input 4	Input 4, PA8 to PA11 Low Voltage

PPC → Option 02.x – Drive Name LV3



CAUTION

- The total electrical load for all connected servo motors **must not** exceed 40VA.

Each Drive connected to the system can be given a name which will then be shown on the text display, there are a maximum of 10 drives (0 – 9), which can be connected, the names are set by option parameter 02.0 to 02.9.

The drives are displayed with the drive number first, then a three character label, such as '1GAS', '2AIR', '3AIR' etc.

The permitted drive names are:

Number entered in parameter 02.X	Drive Name	Considered as:	Trim if selected:	Meaning
0	N/A	Not used	N/A	This drive will not be used.
1	FUE	Fuel drive	YES	Generic fuel drive
2	GAS	Fuel drive	YES	A gas drive
3	OIL	Fuel drive	YES	An oil drive
4	CUP	Fuel drive	YES	Cup speed for rotary cup burners
5	FGR	Fuel drive	NO	Flue Gas Recirculation damper
6	PUM	Fuel drive	YES	VSD controlled oil pump. Can be used for oil warming – see option parameter 9.8
7	WAS	Fuel drive	YES	Waste fuel (combined fuel firing)
8	PRI	Fuel drive	YES	Primary (flame shaping) air
9	FUE	Fuel drive	NO	Generic fuel, but not trimmed.
10	AIR	Air drive	YES	Main combustion air damper
11	FAN	Air drive	YES	Main combustion air fan
12	SEC	Air drive	YES	Secondary air (same as 10 apart from name)
13	AIR	Air drive	NO	Generic Air, but not trimmed.
14	SLE	Air drive	YES	Burner Sleeve

Two (or more) drives can be allocated the same name.

The number entered also tells the control information about the drive. If the number entered here is less than 10, the drive is considered a fuel. If the number is greater than or equal to 10, it is considered an air drive. This only makes a difference when oxygen trim is enabled for a particular drive, since it determines the trim direction. Selection of fuel trim or air trim is made with option parameters 31.1 to 31.4.

Primary air is considered a fuel drive since it is used for flame shaping and should normally track the fuel valve.

PPC → Option 03.x – Drive Serial Number LV3



CAUTION

- The total electrical load for all connected servo motors **must not** exceed 40VA.
- Due to the possibility of High Frequency electrical noise, **BOTH** ends of the shielded signal wire to the VSD **MUST** be grounded

Each Drive connected to the system has a unique serial number allocated to it during production testing. This number cannot be changed and is used to uniquely identify each drive.

When the system is in commissioning mode the serial number of all the connected drives is read by the PPC6000 and displayed as a 'list' to allow the selection of the relevant serial number unit for each drive (0 – 9) to be made.

Scrolling the list in an upward direction will scroll through all drive options available until '0, No Drive' is displayed. Enter this value to disconnect the specified drive from any hardware.

Scrolling the list in a downward direction will stop when the currently selected item is displayed, and the value will stop flashing.

The display will show type information for each drive present (such as the torque rating of a servo), but the commissioning engineer should be aware of which serial number servo is connected to which item on the burner.

If the optional VSD daughterboard is fitted, four additional options are present (reference section 2.6.18):

Displayed Option	Meaning
-a,VSD1:mA	VSD channel 1 is used with a 4-20mA feedback
-b,VSD1:Hz	VSD channel 1 is used with an encoder pulse (frequency) feedback
-c,VSD2:mA	VSD channel 2 is used with a 4-20mA feedback.
-d,VSD2:Hz.	VSD channel 2 is used with an encoder pulse (frequency) feedback.

If a drive serial number is changed, any profiles using that drive will require re-commissioning. A 'profile invalid' fault (see FAULT 64) will lock the burner out on an attempt to fire a profile which has had a previously commissioned drive removed. The profile can be re-commissioned by using the NEXT / ENTER keys in commission ratio mode to verify all points on the combustion curve.

PPC → Option 04.x – Drive Usage Information LV3

The system will allow a maximum of four fuel / air profiles to be entered. Each drive (servo or VSD) can be selected to operate for any, all or none of these profiles. For example: The system can be programmed to use the 'gas' servo on a gas profile – but not an oil profile.

When the control is firing a profile that doesn't use a specific drive that drive can be disconnected or faulty and the control will carry on working. For example, if the gas servo is broken, the burner can be run on oil.

The number to be entered is a binary code:

- +1 if the drive is used for profile 1,
- +2 if the drive is used for profile 2,
- +4 if the drive is used for profile 3,
- +8 if the drive is used for profile 4.

This means that the number entered is from 0 to 15: (see chart below):

Number entered in parameter 4.X	Drive X used for profile(s)
0	NONE
1	1
2	2
3	2+1
4	3
5	3+1
6	3+2
7	3+2+1
8	4
9	4+1
10	4+2
11	4+2+1
12	4+3
13	4+3+1
14	4+3+2
15	4+3+2+1 (ALL)

If these parameters are changed after the burner has been commissioned, any profiles affected will need to be re-commissioned. A 'profile invalid' fault (see FAULT 64) will lock the burner out on an attempt to fire a profile which has had a previously commissioned drive removed.

If a drive has been added / removed from a profile, it is recommended that the specified profile is erased (using option parameter 45.X) before an attempt is made to re-commission it.

Example of use:

Option parameter 4.0 = 4 → drive 0 used for profile 3 only.

Option parameter 4.2 = 1 → drive 2 used for profile 1 only.

Option parameter 4.0 = 15 → drive 0 used for all profiles.

Option parameter 4.4 = 5 → drive 4 used for profile 1 and profile 3 only.

Option 05.x – Drive Options LV3

This parameter is used to specify direction for each servo drive. This parameter has no effect for variable speed drives (VSD). If this parameter is changed, affected profiles must be re-commissioned by a qualified engineer. A 'profile invalid' fault (see FAULT 64) will lock the burner out until this is done.

Number entered in parameter 5.X	Drive Options
0	Drive moves anti-clockwise
1	Drive moves clockwise.

PPC → Options 06.1-06.4 – Profile Naming LV3

This selects the fuel associated with each profile (1 thru 4) to be displayed after the burner-on hours, e.g. 861 hours Profile 1 Gas, visible on the main display during operation.

Normally available options:

Number entered in parameter 6.X	Fuel solenoid valves opened
0	NONE
1	GAS1 + GAS2
2	OIL1
3*	GAS1 + GAS2 + OIL1

*selection 3 allows combined gas and oil firing. This selection may not be available on certain versions of this product.

In addition, for special applications further options may be allowed – again the availability depends of product version.

Options 07.x, 08.x– *RESERVED*

VSD → Option 09.0 - Inverter control accuracy (0 / 1) LV3

This parameter affects the inverter connected to the unit.

Option parameter 09.0 value	Meaning
0	Low accuracy (normal setting). Control will be performed to within approximately 9 display units, $\pm 0.54\text{Hz}$ for a 0-60Hz system.
1	High accuracy. Control will be performed to within approximately 3 display units, $\pm 0.18\text{Hz}$ for a 0-60Hz system.

Note: When using the high accuracy setting, the control may have difficulty with controlling the speed of the inverter. If this occurs, use the low accuracy setting for reliable operation.

VSD → Option 09.1 - Inverter error tolerance (0 / 1) LV3

This option parameter affects the inverter connected to the unit.

Option parameter 09.1 value	Meaning
0	Small tolerance (normal setting). The control will perform a non-volatile lockout if the inverter positioning error exceeds ± 30 for 15 seconds or ± 55 for 3 second (units conform to those seen on the display).
1	Large tolerance. The control will perform a non-volatile lockout if the inverter positioning error exceeds ± 55 for 3 seconds (units conform to those seen on the display).

Note: Only use the large tolerance setting if an inverter error of up to ± 55 will not cause an unsafe combustion condition to occur at any firing position.